Keypads

About

A variety of fibre optic response pads and pointing devices are available for use in imaging suites and testing rooms. These can be used to record volunteer interaction with a stimuli.

These Current Designs systems interface with a PC via a USB connection.

NOTE
The FIL has two optical keypad systems, the newer 932 system and the older Legacy systems.
The Legacy system has been discontinued and should not be used without conversation with the Cognitive Interface team. Legacy systems can be identified as those with braided cables and interfaces with no screen.

User Guide

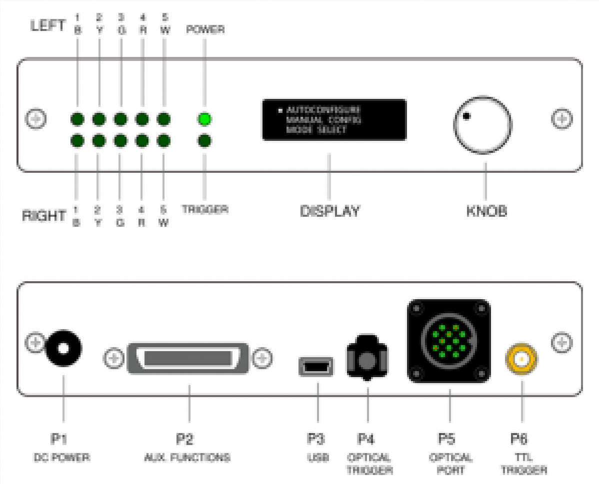
932 Keypad User Guide

To configure an uninstalled standard 932 Keypad system (i.e use in testing room)

follow these instructions

1. **Power the interface unit 'OFF'**
    1. Remove power cable from rear of interface unit, ‘P1’ port
2. **Check all cables are connected** (*a default in scanning suites*)
    1. Ensure the desired keypad is connected to the extension cable
    2. Ensure extension cable is connected to the interface unit, ‘P5’ port
    3. Ensure the interface box is connected to the stimuli presenting PC
        1. Plug USB cable (from ‘P3’ port) into PC
        2. In imaging suites this cable is labelled 'Keypads'
3. **Power the interface unit 'ON'**
    1. Replace power cable to ‘P1’ port
    2. Power indicator LED should be lit
4. **Configure the interface unit settings for the desired set-up**
    1. Check configured keypad
    2. Check keypad outputs
5. **Test keypad output**

    1. Open Notepad/MATLAB to visualise keystrokes
    2. If possible visually check interface lights are lit on keypress
6. **Keypads ready for use**
    1. Fully instruct the volunteer on how to use the keypad
    2. It is recommended to allow volunteers to practice prior to testing
7. **Return to default**
    1. If using different keypads or settings, return to the default when finished



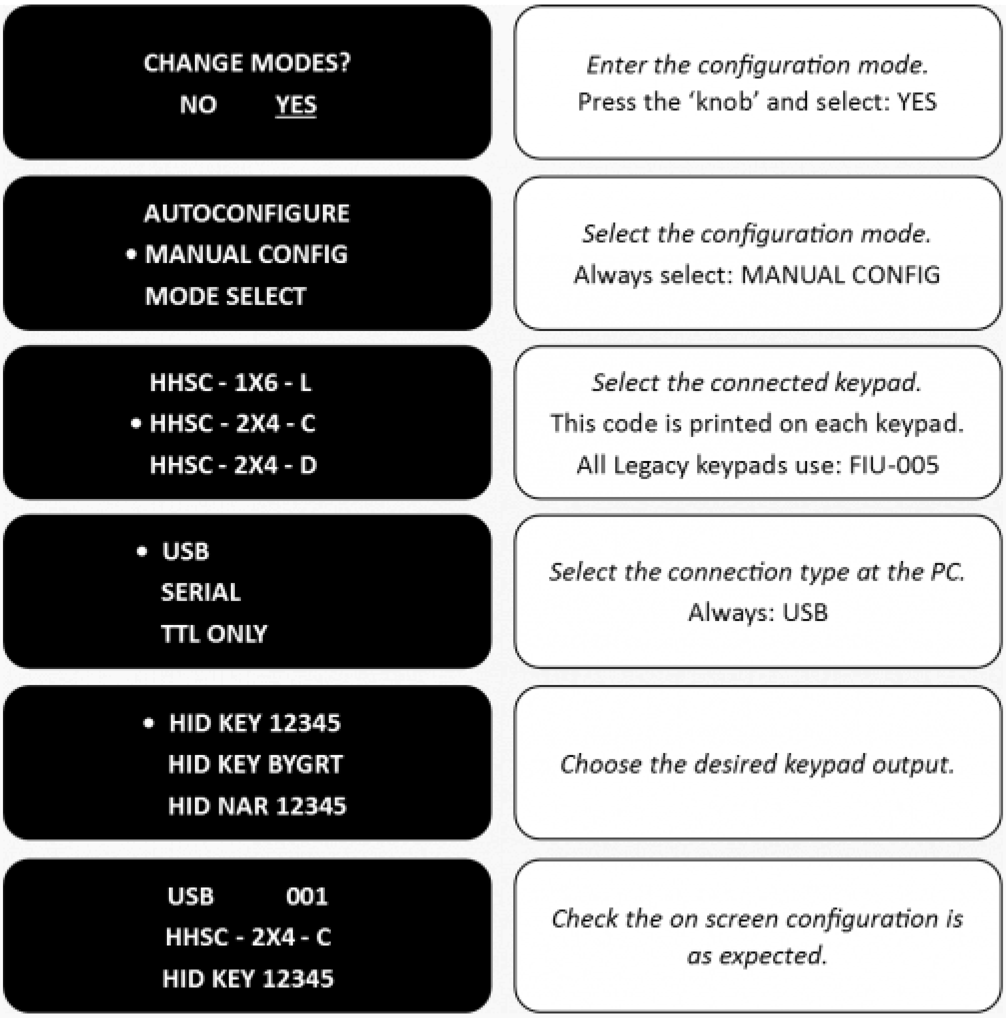
932 Interface Configuration & Keypad Outputs

To configure an installed standard 932 Keypad system used in imaging suites

follow these instructions

Configuration

The 932 interface box will need to be configured to recognise the attached keypad and to produce the desired output.
Follow the below instructions to do this. Rotate the knob to scroll options and press to select.



Keypad Outputs

The 932 and Legacy interface boxes are capable of several different outputs on a keypress. The most common are listed below:

932 Keypad Outputs	
Interface Setting	**Description**
HID KEY BYGRT	Keypress of 'b' for the blue button, 'y' for yellow etc keypress of [need to check output] (left)
HID KEY 1234	Keypress of 1,2,3,4 (right or single) Keypress of 6,7,8,9 (left)
HID NAR BYGRT	Non-Auto Release keypress of 'b' for the blue button, 'y' for yellow etc (right or single) Non-Auto Release keypress of [need to check output] (left)
HID NAR 1234	Non-Auto Release keypress of 1,2,3,4 (right or single) Non-Auto Release keypress of 6,7,8,9 (Left)
FIU-005	If using a Legacy keypad on a 932 interface Right keypress of 1,2,3,4 Left keypress of 6,7,8,9

Manufacturer Resources

Further technical documents and user guides are available on the Current Designs website

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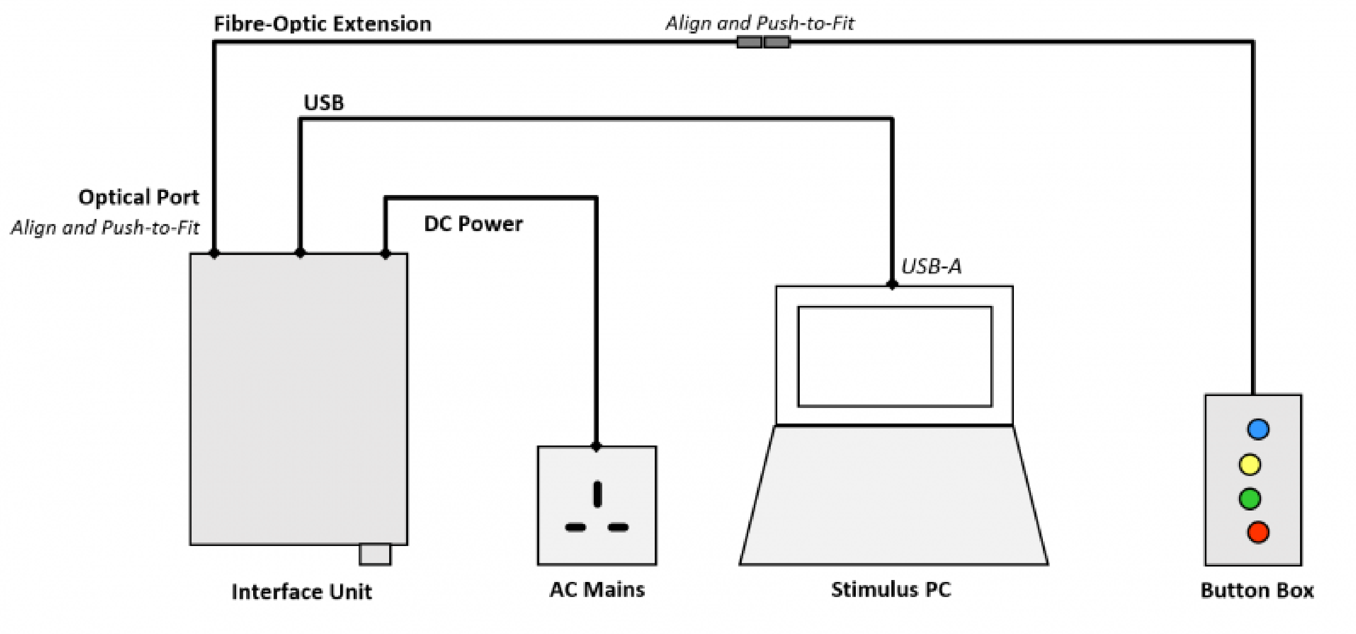
Equipment

932 System

932 System		
Fibre optic system for use in Scanning Suites & Testing Rooms		
	932 Interface Unit	Has a screen and knob for configuration * Power supply * USB cable (micro-USB to USB-A), connecting interface unit to PC
	Extension Bundle	Required for scanning suites.
	Keypads	A variety of keypads are available, contact Cognitive Interface for any requirements

932 Connection Diagram

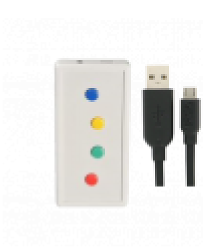
Basic connection diagram for 932 systems below.



932 Keypad Types

There are three types of keypad, available in various button configurations;

- 932 keypad with rubber cable
- USB 4-way keypad, with a USB cable only (no interface box required)
- Legacy keypad with braided cable
    - Do not use these without permission from the Cognitive Interface Team.

8-way (2×4)	4-way (1×4)	5-way (1×5)	4-way (1×4 diamond)	4-way (1×4 USB)
				
932 & Legacy	932 & Legacy	Legacy	932	USB only

USB Keypad System

USB System		
Electrical system for use in Testing Rooms		
	USB Keypads	Keypad connects directly to a PC and does not require an interface box * USB cable (micro-USB to USB-A), connecting keypad to PC

Standard Room Availability

The following keypads are available as standard. Other configurations are available on request.

Modality	Interface Unit	Keypad Type
Quattro MRI	932	8-way (2×4)
Penta MRI	932	8-way (2×4)
7T MRI	932	8-way (2×4)
MEG	932	4-way (1×4)
OPM	932	8-way (2×4)
EEG	932	8-way (2×4)
Testing Rooms	932 or Legacy	Available on request: Legacy interface unit and various keypads USB 4-way (1×4) keypad (Non-MRI)

Script Help

Not all keypads necessarily return the same key codes so best practice would be to make your script flexible and have the script record which keys give which key codes. The script can use those values elsewhere. This practice has the additional advantage that it checks, before your scan, that the key response pad is correctly plugged in and switched on!

See https://github.com/WCHN/PTBexamples for two demos of how to do this in PsychToolbox

Legacy Keypads

Use the following link, for information on the old generation (Legacy) keypad system.
Legacy systems shouldn't be used without approval from the Cognitive Interface Team.

Legacy Systems

- **Legacy (old)**
  - Legacy interface unit
    - AC power supply
    - USB cable (mini-USB to USB-A), connecting interface unit to PC
  - Keypads (braided cable)
  - Extension cable

Legacy System
Old fibre optic system- used for bespoke projects only

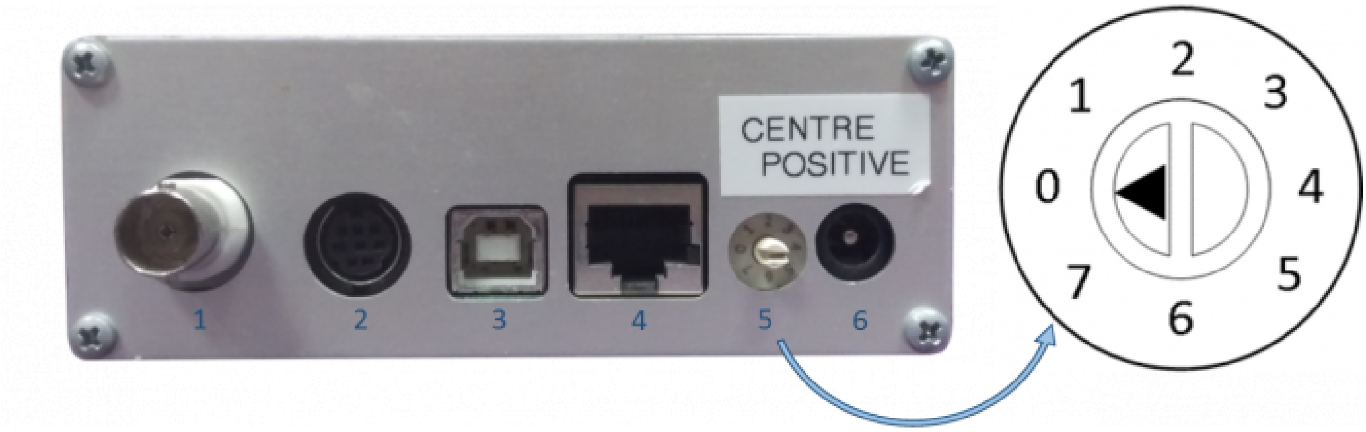
Legacy System		
	Legacy Interface	No screen * AC power supply * USB cable (mini-USB to USB-A), connecting interface unit to PC Primarily reserved for testing room use and spares
	Extension Bundle	Required for scanning suites.
	Keypads	Variety of keypads available list

Legacy Keypad Outputs	
Interface Setting	**Description**
Dial 0 Standard	Keypress of 1,2,3,4 (single 1×4)
Dial 6 8 Button	Keypress of 1,2,3,4 (right) Keypress of 6,7,8,9 (left)
Dial 2 Non-Auto Release	Non-Auto Release keypress of 1,2,3,4 (single 1×4)
Dial 4 Numeric	Not Known
Dial 6 8 Button	Keypress of 1,2,3,4 (right) Keypress of 6,7,8,9 (left)

Legacy Configuration

The Legacy interface box will need to be configured to recognise the attached keypad and to produce the desired output.

1. Use a small screwdriver to turn the dial (port 5) on the rear of the Legacy interface unit.
    1. The number selected on the dial will define the keypad output setting.
    2. Keypad output settings are printed on the underside of the Legacy interface unit.



Troubleshooting

Unexpected or No Keypad Outputs

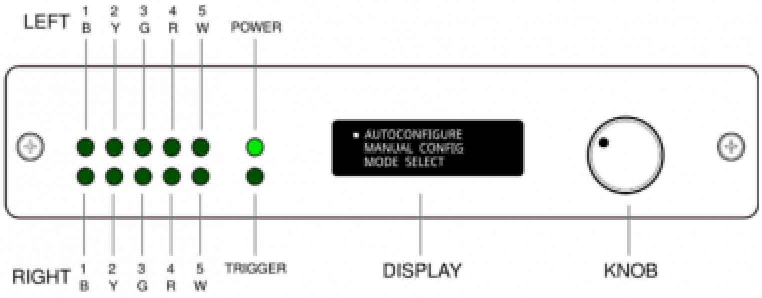
1. Ensure the interface box is connected to the stimuli presenting PC
    1. Plug USB cable (from ‘P3’ port) into PC
    2. In imaging suites this cable is labelled 'Keypads'
2. Check the set interface_configuration_keypad_outputs|configuration]] is correct
3. Power interface unit OFF & ON
    1. Remove power cable from rear of interface unit, ‘P1’ port
    2. Wait 5 seconds
    3. Replace the connection
    4. If easily reached the necessary wall socket could also be switched OFF & ON

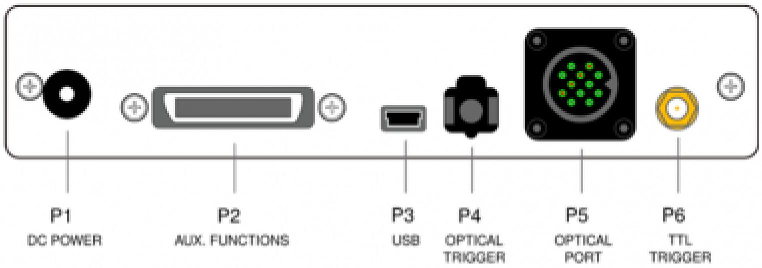
Unwanted Keypad Outputs of ‘5’

1. Ensure the BNC connecter labelled ‘Scanner Pulse’ is not connected to the ‘Synchronisation Box’ attached to the interface unit.
2. Power interface unit OFF & ON
    1. Remove power cable from rear of interface unit, ‘P1’ port
    2. Wait 5 seconds
    3. Replace the connection
    4. If easily reached the necessary wall socket could also be switched OFF & ON

Continuous Keypad Outputs

1. Check the set configuration is correct
2. Power interface unit OFF & ON
    1. Remove power cable from rear of interface unit, ‘P1’ port
    2. Wait 5 seconds
    3. Replace the connection
    4. If easily reached the necessary wall socket could also be switched OFF & ON





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